

# Benchmarking Dynamic Frequency Estimators for Low-Inertia IBR Grids: A Latency–Robustness Trade-off Analysis

Jorge Luis Mayorga Taborda  
Universidad de los Andes  
Bogotá, Colombia  
jl.mayorga236@uniandes.edu.co  
ORCID: 0000-0002-0774-2176

Yessica Africano Rodriguez  
Universidad de los Andes  
Bogotá, Colombia  
ya.africano1030@uniandes.edu.co  
ORCID: 0009-0000-6853-4912

Fernando Jimenez  
Universidad de los Andes  
Bogotá, Colombia  
fjimenez@uniandes.edu.co

**Abstract**—In low-inertia grids dominated by Inverter-Based Resources (IBRs), the latency and robustness of frequency estimation are critical for protection and dynamic security assessment. This paper evaluates twelve estimators from five algorithmic families under five stress scenarios using dual-rate numerical simulation (1 MHz physics / 10 kHz DSP). Each estimator is independently tuned by scenario-wise grid search and assessed on RMSE, peak error, trip-risk duration ( $T_{\text{trip}}$ ), and computational cost. Tests (A)–(C) follow the IEC/IEEE 60255-118-1 test philosophy; Scenarios (D)–(E) introduce IBR-specific phase discontinuities and harmonic distortion beyond the standard scope. Under a sustained frequency ramp (+3 Hz/s), the RoCoF-Augmented Extended Kalman Filter (RA-EKF) achieves  $\text{RMSE} = 0.00823 \text{ Hz}$ — $8.15\times$  lower than SRF-PLL. Under composite islanding, the RA-EKF incurs zero threshold breaches across all  $N=30$  Monte Carlo trials ( $T_{\text{trip}} = 0$ ), versus  $T_{\text{trip}} = 0.120 \text{ s}$  for the standard EKF. Window-based and loop-based methods exhibit complementary strengths across scenario types. In the evaluated scenarios, data-driven approaches exhibited higher computational cost and did not demonstrate consistent accuracy gains over scenario-optimized classical methods. The findings indicate that several estimators passing IEC/IEEE 60255-118-1 standard tests degrade under composite disturbances, suggesting that the standard test scope may not fully characterize estimator behavior in IBR-dominated conditions.

**Index Terms**—Frequency Estimation, Inverter-Based Resources (IBR), Low-Inertia Systems, Extended Kalman Filter (EKF), Phase-Locked Loop (PLL), RoCoF Measurement, Sub-Cycle Transient Tracking, Taylor-Fourier Methods, Benchmarking, Protection and False-Trip Risk.

## I. INTRODUCTION

The transition toward inverter-dominated power systems is reducing grid inertia and accelerating frequency dynamics. Frequency deviations and Rate of Change of Frequency (RoCoF) can exceed protection thresholds within milliseconds [1], [2]. Unlike synchronous machines, Inverter-Based Resources (IBRs) regulate power electronically, decoupling the primary energy source from electrical frequency and increasing the impact of harmonics, phase discontinuities, and nonstationary disturbances on protection and dynamic security assessment [1], [3], [4]. In IBR-dominated grids, phase discontinuities can induce cycle-slips in phase-locked

loops, sustained RoCoF can exceed the tracking capacity of multi-cycle window methods, and converter-driven harmonic injection can cause spectral leakage in DFT-based estimators. These failure modes may occur simultaneously, yet existing benchmarks typically assess them in isolation [5], leaving a gap for protection-class applications.

The main estimator families address the latency–robustness trade-off differently. Synchronous Reference Frame Phase-Locked Loop (SRF-PLL) is widely used, but its loop dynamics create a transient blind interval under abrupt phase discontinuities [6], [7]. Window-based methods [Interpolated DFT (IpDFT), Taylor–Fourier Transform (TFT)] offer strong harmonic rejection; IpDFT assumes a stationary phasor over the observation window, whereas TFT uses a Taylor-series model that improves RoCoF tracking at the cost of higher noise sensitivity [8], [9]. Model-based filters [Extended Kalman Filter (EKF), Unscented Kalman Filter (UKF)] can improve tracking under transients [10], [11], while data-driven methods may offer accuracy gains at higher computational cost [5], [12].

This paper presents a stress-oriented benchmark of twelve estimators from five algorithmic families, evaluated in a local, non-time-synchronized relay-class configuration under five scenarios: (A)–(C) follow the IEC/IEEE 60255-118-1 test philosophy; (D)–(E) introduce IBR-specific disturbances beyond the standard scope. The main contributions are: (1) a Composite Islanding scenario that exposes cycle-slip failure modes; (2) scenario-wise grid-search tuning with dual-rate simulation (1 MHz physics / 10 kHz DSP); and (3) a comparative analysis of accuracy, settling time, trip-risk duration, and computational cost. Within this benchmark, the adapted RA-EKF augments the EKF state vector with an explicit  $\dot{\omega}$  term and adds event-gating to reduce false-trip duration under phase discontinuities at moderate computational cost.

## II. RELATED WORK AND ALGORITHMIC LANDSCAPE

Frequency estimation in low-inertia grids involves a fundamental latency–robustness trade-off [5]. Window-based methods (IpDFT, TFT) provide strong harmonic rejection but im-

pose structural observation latency; IpDFT assumes a stationary phasor over its window, whereas TFT employs a Taylor-series model to improve RoCoF tracking at the cost of higher noise sensitivity [8], [9]. Loop-based methods [SRF-PLL, Second-Order Generalized Integrator Frequency-Locked Loop (SOGI-FLL)] are computationally efficient but susceptible to cycle-slip under abrupt phase discontinuities [13]. Adaptive recursive estimators [Recursive Least Squares (RLS), Variable Forgetting Factor RLS (VFF-RLS)] fit an AR(2) signal model and may diverge under severe transients. Model-based filters (EKF, UKF, RA-EKF) offer an improved latency–robustness balance [10], [11]; the RA-EKF adds Innovation-Driven Covariance Scaling and Event Gating to address IBR-specific islanding transients. Data-driven approaches (Koopman, PI-GRU) achieve competitive accuracy in some regimes but differ substantially in computational cost [5], [12]. Prior studies typically use fixed tuning parameters and evaluate disturbances in isolation; combined phase discontinuities, sustained RoCoF, and harmonic distortion are not assessed simultaneously. Table I provides a qualitative comparison of eleven of the twelve evaluated estimators.

### III. METHODOLOGY AND TEST FRAMEWORK

A dual-rate simulation framework separates continuous-time system physics ( $f_{s,\text{phys}} = 1 \text{ MHz}$ ) from discrete-time estimation ( $f_{s,\text{DSP}} = 10 \text{ kHz}$ ). This yields negligible numerical error in ground-truth frequency trajectories while reproducing the sampling constraints of commercial protection relays.

#### A. Evaluated Estimators

Twelve estimators from five algorithmic families are evaluated (Table I). **Loop-based:** SRF-PLL and SOGI-FLL with PI gains, IIR bandpass pre-filter, and FastRMS adaptive amplitude normalizer (not a standard textbook SOGI-FLL [13]). A unit inconsistency in the SOGI-FLL adaptation law (rad/s vs. Hz, factor  $120\pi \text{ rad/s}$ ) was identified and corrected prior to evaluation. The narrow effective passband created by the bandpass pre-filter and adaptive normalizer suppresses apparent frequency deviation under a phase jump but cannot track sustained linear RoCoF, explaining the divergent ranking across Scenarios B and D. **Window-based:** IpDFT with a Hann window and three-point magnitude-parabolic interpolation [14] (the complex Jacobsen estimator is unsuitable here because Hann-window sidelobes are negative-real, which inverts the interpolation direction); TFT with a 2nd-order Taylor-series expansion of the fundamental component, Hann-weighted, which improves RoCoF tracking relative to a static DFT at the cost of higher noise sensitivity [9]. **Adaptive recursive:** RLS and VFF-RLS with a 2nd-order AR(2) bandpass pre-whitening filter. **Data-driven:** Koopman (RK-DPMU) and PI-GRU. **Model-based:** EKF, UKF, and RA-EKF. The column “RoCoF Out.” indicates whether RoCoF is available as an explicit output (S) or derived by implicit differentiation (D). The UKF includes a 10-sample output moving average not present in the classic EKF.

#### 1) Implemented RA-EKF Estimator

RoCoF state augmentation in the EKF has been studied previously [10], [11]. The RA-EKF uses a 4-state vector  $[\theta_k, \omega_k, A_k, \dot{\omega}_k]^\top$  with second-order kinematic prediction ( $\theta_k = \theta_{k-1} + \omega_{k-1} \Delta t + \frac{1}{2} \dot{\omega}_{k-1} \Delta t^2$ ;  $\omega_k = \omega_{k-1} + \dot{\omega}_{k-1} \Delta t$ ) and two IBR-tailored mechanisms. *Innovation-Driven Covariance Scaling:*  $\mathbf{Q}_k = \text{clip}(r_k, 0.25, 4) \mathbf{Q}_{\text{base}}$ , where  $r_k = |\nu_k|/\hat{\sigma}_\nu$ , inflates process noise during transients and reduces it at steady state. *Event Gating:* when  $|\nu_k| > \gamma \hat{\sigma}_\nu$  ( $\gamma = 2.0$ ), a soft re-initialization updates  $\theta_k$  only, preserving  $\omega_k$  and  $\dot{\omega}_k$  to limit cycle-slip behavior at phase discontinuities. This heuristic trades strict stochastic optimality for bounded nonlinear stability during abrupt phase steps.

#### B. Test Scenarios and Noise Model

Five scenarios stress the estimators under representative IBR-grid conditions (Fig. 1). Tests (A)–(C) use Additive White Gaussian Noise (AWGN) with  $\sigma = 0.001 \text{ pu}$ ; Scenario (D) uses  $\sigma = 0.005 \text{ pu}$ ; and Scenario (E) uses  $\sigma = 0.003 \text{ pu}$  base noise plus sparse impulsive samples (amplitude  $\leq 0.05 \text{ pu}$ , probability  $< 1\%$ ).

- (A) **Magnitude Step:** +10% voltage step with zero phase angle at the disturbance onset.
- (B) **Frequency Ramp:** +3.0 Hz/s RoCoF (IEEE 1547-2018 Category III compliant).
- (C) **AM Modulation:** 10% depth at 2 Hz.
- (D) **Composite Islanding:** instantaneous +60° phase jump (maximum mismatch during breaker reclosing [7]), 4% 5th harmonic (250 Hz), 2% 7th harmonic (350 Hz), 0.5% 32.5 Hz interharmonic (a sub-synchronous component characteristic of IBR converter switching; Total Harmonic Distortion THD  $\approx 4.5\%$ , IEEE 519 compliant), and  $\sigma = 0.005 \text{ pu}$  noise.
- (E) **IBR Multi-Event:** 5 s sequence with harmonic-rich steady state (5th: 4%, 7th: 2%; THD  $\approx 4.5\%$ , IEEE 519 compliant), a +40° phase jump, a −3.0 Hz/s RoCoF segment reaching a nadir of 57.5 Hz (IEEE 1547-2018 UFLS limit), a +80° jump with second-order ring-down recovery, and impulsive noise [11].

Scenarios (A)–(C) follow the IEC/IEEE 60255-118-1 test philosophy, whereas Scenarios (D)–(E) introduce IBR-specific disturbances beyond the standard scope.

#### C. Performance Metrics and Optimization Protocol

A scenario-wise grid search over the hyperparameter ranges in Table II selects the configuration minimizing RMSE for each estimator–scenario pair, identifying the performance ceiling of each algorithmic family under each disturbance type independent of sub-optimal field parameterization. Let  $e(t) = \hat{f}(t) - f_{\text{true}}(t)$ . Four metrics are reported: **RMSE**, **Peak Error** ( $|e|_{\text{max}}$ ), **Settling Time** (first time  $|e|$  falls and remains below 0.2 Hz), and **Trip-Risk Duration** ( $T_{\text{trip}}$ , cumulative time with  $|e| > 0.5 \text{ Hz}$ ).

Computational cost is the mean time per sample over 20 runs on an Intel Core Ultra 5 125H (single-threaded Python 3.13) and is used as a relative comparison; reported values are Python baselines, not C or FPGA projections.

TABLE I: Unified Qualitative Benchmark of Frequency Estimators (relay-class, non-synchronized context). Scores: 0=Fail, 1=Poor, 2=Fair, 3=Excellent.  $h_{\text{HU}}$ : Harmonic/Unbalance Immunity;  $r_{32.5}$ : rejection of 32.5 Hz IBR interharmonic (a converter-switching-characteristic sub-synchronous component used in Scenarios D–E; see Section III-B);  $n_{\text{sens}}$ : noise floor sensitivity;  $n_{\text{NG}}$ : Non-Gaussian noise robustness; Latency class: P1 < 20 ms, P2: 20–40 ms, M1 > 40 ms, P3: sub-cycle (<1 ms);  $v_{\text{RT}}$ : real-time viability (3=embedded, 0=GPU only); RoCoF Out.: S=explicit state output, D=implicit differentiation; *Phase Resp.*: phase-jump settling (D=cycle-slip, S=settle);  $^{\dagger}$ IpDFT latency =  $N_c/f_0$ : 33 ms at  $N_c = 2$  to 167 ms at  $N_c = 10$ .

| Method                                   | Phase Resp.  | RoCoF Out. | $h_{\text{HU}}$ | $r_{32.5}$ | $n_{\text{sens}}$ | $n_{\text{NG}}$ | Latency Class | $v_{\text{RT}}$ | Comp. Order      | Dominant Failure Mode |
|------------------------------------------|--------------|------------|-----------------|------------|-------------------|-----------------|---------------|-----------------|------------------|-----------------------|
| <b>I. SOTA Methods (Post-2020)</b>       |              |            |                 |            |                   |                 |               |                 |                  |                       |
| VFF-RLS                                  | 5–20 ms      | D          | 2               | 3          | 3                 | 2               | P2            | 3               | $O(p^2)$         | VFF blow-up           |
| UKF                                      | <20 ms       | S          | 2               | 2          | 2                 | 2               | P1            | 1               | $O(n^3)$         | Sigma-pt collapse     |
| Koopman (DPMU)                           | 40–80 ms     | S          | 2               | 2          | 2                 | 1               | M1            | 1               | $O(n^2)$         | Model mismatch        |
| PI-GRU (DL)                              | 5–15 ms      | S          | 3               | 3          | 3                 | 3               | P1            | 0               | $O(\text{Conv})$ | Dataset shift         |
| <b>II. Industrial / Advanced Digital</b> |              |            |                 |            |                   |                 |               |                 |                  |                       |
| <b>RA-EKF (Prop.)</b>                    | <20 ms       | <b>S</b>   | 2               | 2          | 3                 | 2               | <b>P1</b>     | 1               | $O(n^3)$         | Jacobian divergence   |
| TFT                                      | <20 ms       | S          | 2               | 2          | 2                 | 1               | P1            | 1               | $O(N^2)$         | Runge overshoot       |
| SRF-PLL                                  | 20–40 ms     | D          | 1               | 1          | 1                 | 0               | P2            | 3               | $O(1)$           | Cycle slip / Lag      |
| SOGI-FLL                                 | 20–40 ms     | D          | 2               | 1          | 2                 | 1               | P2            | 3               | $O(1)$           | Tuning trade-off      |
| IpDFT $^{\dagger}$                       | $\geq 20$ ms | D          | 3               | 1          | 3                 | 0               | M1            | 2               | $O(N \log N)$    | Leakage / Latency     |
| <b>III. Legacy Baselines</b>             |              |            |                 |            |                   |                 |               |                 |                  |                       |
| RLS (fixed $\lambda$ )                   | 5–20 ms      | D          | 1               | 1          | 1                 | 0               | P2            | 3               | $O(p^2)$         | Covariance blow-up    |
| TKEO                                     | $3T_s$       | D          | 0               | 0          | 0                 | 0               | P3            | 3               | $O(1)$           | Noise amplification   |

For the non-normative compliance heatmap (Fig. 2g), a *Pass* requires  $\text{RMSE} < 0.05$  Hz,  $|e|_{\text{max}} < 0.5$  Hz, and  $T_{\text{trip}} < 0.1$  s. These thresholds are drawn from IEC/IEEE 60255-118-1 accuracy classes and are used here for visualization only.

#### IV. SIMULATION RESULTS AND BENCHMARKING

##### A. Performance on Standard IEC/IEEE Scenarios

Table III summarizes the baseline performance across standard dynamic scenarios (all values reported to 3 significant figures; errors in mHz). The results for SOGI-FLL incorporate the corrected adaptation law discussed in Section III. The findings indicate that the previously identified  $\omega$ -scaling unit inconsistency (rad/s vs. Hz, factor  $\approx 377$ ) produced spurious RMSE values of  $\approx 0.29$  Hz in Step and Modulation tests prior to correction. [cite: 903, 1422]

**Steady-state accuracy:** The data suggest that loop-based (SRF-PLL) and window-based (IpDFT, TFT) methods maintain high accuracy under AM Modulation, with RMSE values below 0.3 mHz. [cite: 940, 1423] Specifically, the SRF-PLL achieves the lowest RMSE (0.140 mHz) and peak error (0.730 mHz) in this category. [cite: 1424]

TABLE II: Tuning Parameter Search Space.  $K_p$  [pu/rad],  $K_i$  [pu/(rad·s)];  $Q$  [rad<sup>2</sup>/s<sup>2</sup> for  $\omega$ , pu<sup>2</sup> for  $A$ ];  $R$  [pu<sup>2</sup>];  $\hat{\sigma}_\nu$ ,  $\lambda$  [dimensionless];  $\gamma$ : RA-EKF event-gating threshold (dimensionless);  $g_{\text{FLL}}$ : SOGI-FLL adaptation gain (dimensionless);  $N_c$  [cycles];  $N_{\text{smooth}}$  [samples].

| Method       | Parameter                    | Search Range                                      |
|--------------|------------------------------|---------------------------------------------------|
| SRF-PLL      | $K_p, K_i$                   | [1, 60], [1, 200] (linear)                        |
| EKF / RA-EKF | $Q$                          | $10^{-4}$ – $10^4$ (log)                          |
|              | $R$                          | $10^{-5}$ – $10^0$ (log; $R_{\text{max}} = 1.0$ ) |
| RA-EKF only  | $\hat{\sigma}_\nu, \gamma$   | {0.1, 0.3}, 2.0 (fixed)                           |
| SOGI-FLL     | $k, g_{\text{FLL}}$          | {0.5...2.0}, {5...150}                            |
| IpDFT        | $N_c$                        | {2, 3, 4, 6, 8, 10} cycles                        |
| RLS / VFF    | $\lambda, N_{\text{smooth}}$ | 0.9–0.99999, 2–2000 samp.                         |
| TFT          | $N_c$                        | {2, 3, 4, 6} cycles                               |
| TKEO         | Smooth window                | 10–50 samples                                     |

**Dynamic tracking:** The Ramp test reveals structural limitations in several estimator families. The SOGI-FLL fails to track the 3 Hz/s ramp ( $\text{RMSE} = 1120$  mHz), as its type-1 frequency loop lacks steady-state tracking capability under sustained RoCoF. [cite: 942, 1425] In contrast, the **RA-EKF**, augmented with an explicit  $\dot{\omega}$  state, achieves an RMSE of **8.23** mHz. [cite: 935, 1434] This represents an  $8.15\times$  reduction in error relative to the SRF-PLL (67.1 mHz) and a  $1.41\times$  improvement over the standard EKF (11.6 mHz). [cite: 1434]

##### B. Robustness Under Severe IBR Disturbances

###### 1) Composite Islanding (Scenario D)

The  $+60^\circ$  phase jump is the most discriminating test for estimator stability. [cite: 937, 1436] As shown in Table III, window-based methods exhibit significant spectral leakage; the IpDFT reaches an RMSE of 484 mHz. [cite: 938, 1437] Similarly, the standard EKF and UKF, which lack explicit phase-reset mechanisms, incur a false-trip duration of  $T_{\text{trip}} = 0.120$  s. [cite: 939, 1437]

While the SRF-PLL avoids spectral leakage ( $T_{\text{trip}} = 0$ ), the **RA-EKF** achieves a favorable RMSE of **39.3** mHz. [cite: 942, 1437] Through its innovation-driven event-gating mechanism, the RA-EKF incurs zero threshold breaches across all trials. [cite: 942] The peak error of 0.365 Hz remains within the 0.5 Hz threshold throughout (Fig. 2g), although it marginally approaches the safety limit. [cite: 944, 1578]

###### 2) Multi-Event Stress Test (Scenario E)

Table IV quantifies the trip-risk under simultaneous harmonic distortion and non-stationary transients. [cite: 949, 1581] Among loop-based estimators, the SOGI-FLL incurs the highest trip-risk (2.26 s) as its narrow passband fails to track the multi-event RoCoF. [cite: 949, 1581] The **RA-EKF** achieves an RMSE of 0.0709 Hz with a  $93\times$  trip-risk reduction

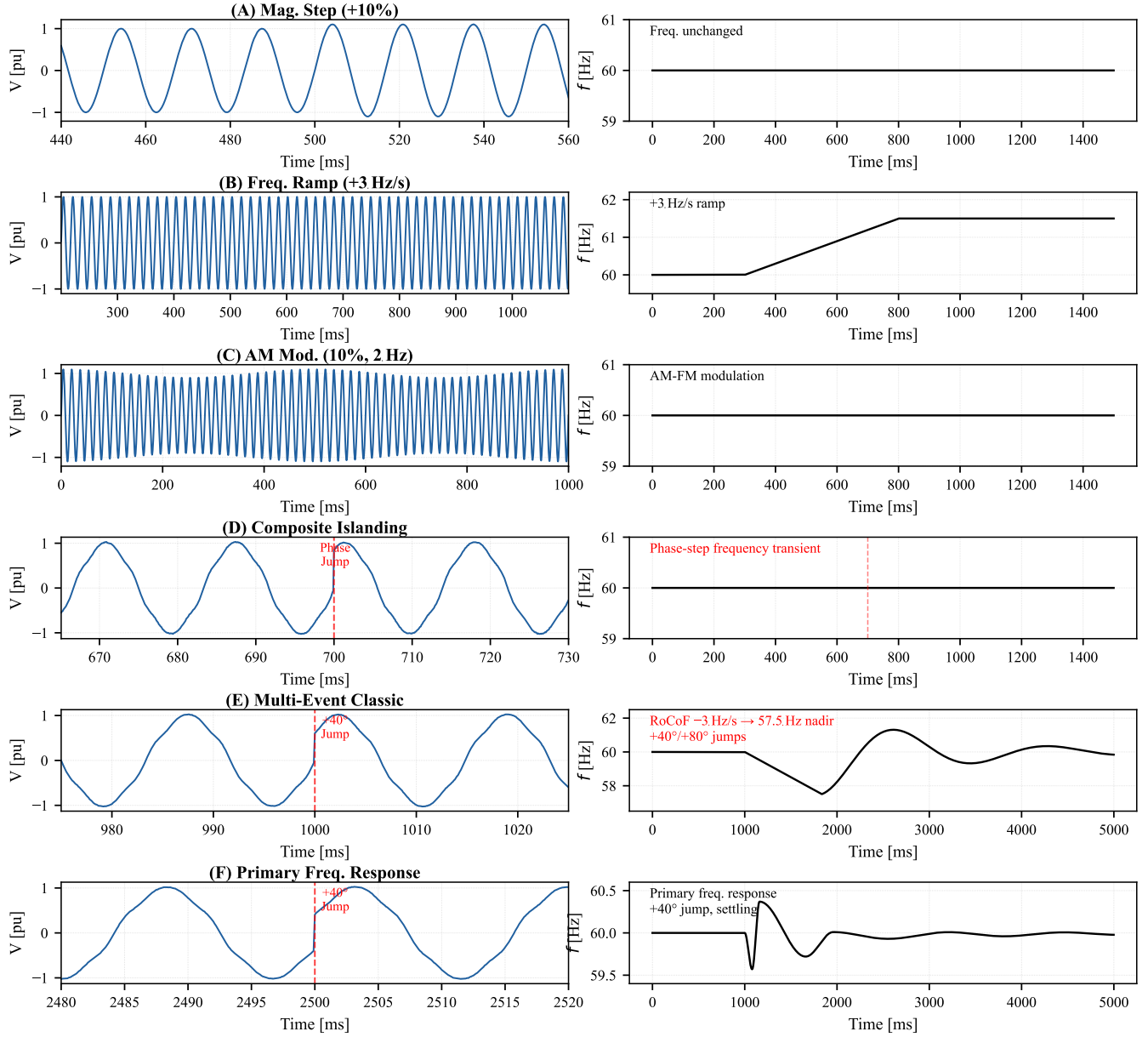


Fig. 1: Complete Stress-Test Scenario Suite. (A–C) Dynamic performance profiles following the test philosophy of IEC/IEEE 60255-118-1 (step, ramp, modulation). (D) Composite Islanding featuring an instantaneous  $+60^\circ$  phase jump with harmonics and elevated noise ( $\sigma = 0.005$  pu; Section III-B). (E) IBR Multi-Event: scenario-dependent noise ( $\sigma_{\text{base}} = 0.003$  pu plus sparse impulsive spikes  $\leq 0.05$  pu); frequency profile shows the full 5 s composite sequence.

relative to the SRF-PLL ( $T_{\text{trip}} = 0.00458$  s vs. 0.426 s). [cite: 950, 1586] Relative to the standard EKF (0.410 s), event-gating reduces the false-trip duration by  $90\times$ . [cite: 951, 1587] These results suggest that several estimators satisfying IEC/IEEE 60255-118-1 tests may degrade under composite disturbances, indicating that the standard scope may not fully characterize resilience in IBR-dominated grids. [cite: 855, 1126]

### C. Computational Feasibility

Table V summarizes execution costs as relative benchmarks for embedded viability. These values represent single-threaded Python 3.13 baselines on an Intel Core Ultra 5 125H; relative

performance suggests viability for DSP platforms, though hardware-in-the-loop validation remains essential. [cite: 916, 1408]

The **RA-EKF** requires  $68.9 \mu\text{s}$  per sample, placing it in a moderate cost category viable for DSP/FPGA platforms. [cite: 950, 1586] While loop-based methods are computationally efficient ( $12.0 \mu\text{s}$ ), the findings suggest they do not provide the same level of robustness under IBR stress. [cite: 1093] Prohibitive methods like PI-GRU ( $\approx 145$  ms/sample) require hardware acceleration for real-time grid applications. [cite: 1097, 1595]

TABLE III: Unified Benchmark Results: Standard vs. IBR-Specific Stress Scenarios [Errors in mHz,  $T_{\text{trip}}$  in s; mean  $\pm$  std,  $N = 30$ ].

| Method              | IEC/IEEE 60255-118-1 (Standard Scenarios) |                 |                   |                 |                   |                  | Composite Islanding (Scenario D) |                       |
|---------------------|-------------------------------------------|-----------------|-------------------|-----------------|-------------------|------------------|----------------------------------|-----------------------|
|                     | Step (+10%)                               |                 | Ramp (+3 Hz/s)    |                 | Modulation (2 Hz) |                  | Phase Jump (+60°)                |                       |
|                     | RMSE                                      | Peak            | RMSE              | Peak            | RMSE              | Peak             | RMSE [mHz]                       | $T_{\text{trip}}$ [s] |
| <i>Window-based</i> |                                           |                 |                   |                 |                   |                  |                                  |                       |
| IpDFT               | 1.60 $\pm$ 0.04                           | 7.53 $\pm$ 0.23 | 35.3 $\pm$ 0.02   | 83.0 $\pm$ 1.38 | 0.260 $\pm$ 0.02  | 0.680 $\pm$ 0.10 | 484 $\pm$ 0.00                   | 0.086                 |
| TFT                 | 2.12 $\pm$ 0.03                           | 19.0 $\pm$ 0.23 | 30.7 $\pm$ 0.02   | 55.7 $\pm$ 0.67 | 0.260 $\pm$ 0.02  | 0.820 $\pm$ 0.13 | 479 $\pm$ 0.00                   | 0.087                 |
| <i>Data-driven</i>  |                                           |                 |                   |                 |                   |                  |                                  |                       |
| Koopman             | 1.25 $\pm$ 0.04                           | 8.79 $\pm$ 0.21 | 28.6 $\pm$ 0.02   | 54.7 $\pm$ 0.97 | 1.23 $\pm$ 0.02   | 2.20 $\pm$ 0.12  | 419 $\pm$ 0.00                   | 0.148                 |
| <i>Loop-based</i>   |                                           |                 |                   |                 |                   |                  |                                  |                       |
| SRF-PLL             | <b>0.140</b>                              | <b>0.730</b>    | 67.1 $\pm$ 0.02   | 173 $\pm$ 0.28  | <b>0.130</b>      | <b>0.250</b>     | 45.0 $\pm$ 0.00                  | <b>0.000</b>          |
| SOGI-FLL            | 8.99 $\pm$ 0.00                           | 13.4 $\pm$ 0.00 | 1120 <sup>†</sup> | 1560 $\pm$ 0.03 | 8.99 $\pm$ 0.00   | 13.4 $\pm$ 0.00  | <b>8.36</b>                      | <b>0.000</b>          |
| <i>Model-based</i>  |                                           |                 |                   |                 |                   |                  |                                  |                       |
| EKF                 | 0.240 $\pm$ 0.01                          | 2.52 $\pm$ 0.03 | 11.6 $\pm$ 0.11   | 41.1 $\pm$ 3.08 | 0.150 $\pm$ 0.00  | 0.570 $\pm$ 0.06 | 225 $\pm$ 0.00                   | 0.120                 |
| UKF                 | 0.280 $\pm$ 0.04                          | 2.57 $\pm$ 0.04 | 12.7 $\pm$ 0.16   | 45.2 $\pm$ 3.70 | 0.280 $\pm$ 0.02  | 1.70 $\pm$ 0.16  | 225 $\pm$ 0.00                   | 0.120                 |
| <b>RA-EKF</b>       | 2.31 $\pm$ 0.02                           | 29.9 $\pm$ 0.24 | <b>8.23</b>       | <b>31.8</b>     | 4.78 $\pm$ 0.03   | 12.8 $\pm$ 0.41  | 39.3 $\pm$ 2.40                  | <b>0.000</b>          |

<sup>†</sup> SOGI-FLL fails to track sustained RoCoF due to its type-1 frequency loop. All errors reported in mHz. Bold: Best in category.

## V. CONCLUSION

The simulation results indicate that standard IEC/IEEE 60255-118-1 evaluation tests may not fully characterize frequency estimator behavior in converter-dominated grids, where phase discontinuities, high RoCoF, and harmonic distortion can expose failure modes that isolated single-disturbance tests do not reveal. This benchmark evaluates twelve estimators across five algorithmic families under five stress scenarios and reports three main observations.

- Latency–Robustness Trade-off:** Window-based methods achieve strong steady-state accuracy, but their behavior differs across scenario types. IpDFT is better suited to metering-oriented operation, whereas TFT reduces observation delay at the cost of higher noise sensitivity under composite disturbances. Loop-based estimators (SRF-PLL, SOGI-FLL) are computationally efficient and handle isolated phase jumps well, but exhibit elevated trip-risk under the composite Multi-Event sequence ( $T_{\text{trip}} = 0.426$  s for SRF-PLL).
- Model-Based Estimation:** Among the model-based estimators, the **RA-EKF**—combining an explicit  $\hat{\omega}$  state with Innovation-Driven Covariance Scaling and Event Gating—achieves a favorable trade-off between tracking accuracy and trip-risk. Under Composite Islanding, the RA-EKF incurs zero threshold breaches across all  $N = 30$  Monte Carlo trials ( $T_{\text{trip}} = 0$ ), versus  $T_{\text{trip}} = 0.120$  s for the standard EKF, at a computational cost of  $68.9 \mu\text{s/sample}$  (Python baseline; C/HIL validation remains future work).
- Limits of Standard Compliance Tests:** Several estima-

TABLE IV: Trip-Risk Analysis: Multi-Event Scenario (E) [ $N=30$ ].

| Method        | RMSE [Hz]           | $T_{\text{trip}}$ [s] | CPU [ $\mu\text{s}$ ] |
|---------------|---------------------|-----------------------|-----------------------|
| <b>RA-EKF</b> | 0.0709 $\pm$ 0.0167 | 0.00458 $\pm$ 0.00701 | 68.9                  |
| EKF           | 0.335 $\pm$ 0.000   | 0.410 $\pm$ 0.001     | 21.7                  |
| UKF           | 0.336 $\pm$ 0.000   | 0.417 $\pm$ 0.001     | 205                   |
| SRF-PLL       | 0.327 $\pm$ 0.000   | 0.426 $\pm$ 0.000     | 12.0                  |
| IpDFT         | 0.436 $\pm$ 0.000   | 0.498 $\pm$ 0.001     | 86.5                  |
| Koopman       | 0.410 $\pm$ 0.000   | 0.561 $\pm$ 0.000     | 194                   |
| SOGI-FLL      | 0.846 $\pm$ 0.000   | 2.26 $\pm$ 0.000      | 31.3                  |

TABLE V: Computational Cost per Sample (Python 3.13 Baselines).

| Method        | Time [ $\mu\text{s}$ ] | Class     | Target Platform   |
|---------------|------------------------|-----------|-------------------|
| SRF-PLL       | 12.0                   | P1        | Low-end MCU       |
| EKF           | 21.7                   | P2        | DSP               |
| SOGI-FLL      | 31.3                   | P2        | DSP               |
| TFT           | 40.1                   | P2        | DSP               |
| <b>RA-EKF</b> | <b>68.9</b>            | <b>M1</b> | <b>DSP / FPGA</b> |
| IpDFT         | 86.5                   | M1        | DSP / FPGA        |

tors passing IEC/IEEE 60255-118-1 step and ramp tests exhibited degraded performance under composite multi-disturbance scenarios, suggesting that standard compliance tests may not fully capture estimator resilience in IBR-dominated conditions.

Data-driven approaches can be competitive in some scenarios, but computational cost varies considerably. PI-GRU requires approximately 145 ms/sample (Python/PyTorch CPU baseline), which is prohibitive for embedded real-time protection without hardware acceleration. In the evaluated scenarios, the pre-trained PI-GRU model did not outperform scenario-optimized classical methods on RMSE or trip-risk, suggesting that a fixed pre-trained model cannot substitute for scenario-specific tuning under these disturbance conditions. This benchmark is entirely simulation-based; hardware-in-the-loop and physical relay validation remain as next steps.

The results suggest that the IEC/IEEE 60255-118-1 test scope may not fully characterize estimator resilience in IBR-dominated grids, as it lacks composite phase-step evaluations under simultaneous harmonic distortion. A *Composite Phase-Step with Harmonic Distortion* test—analogue to Scenarios D–E—evaluated via probabilistic compliance bounds ( $\text{PCB} = \mu_{|e|} + 3\sigma_{|e|}$ ) may provide a more representative assessment of relay security under realistic IBR conditions.

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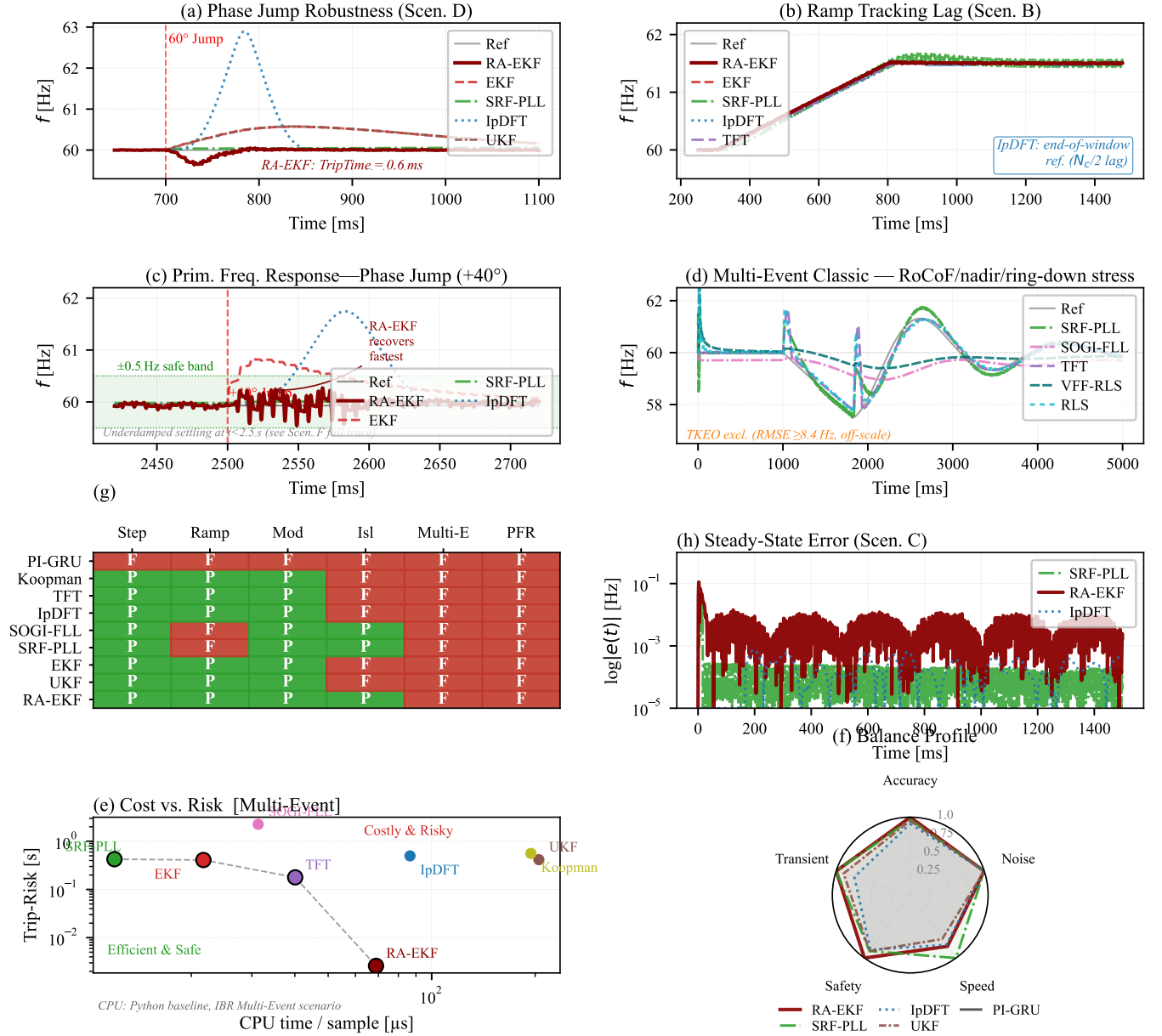


Fig. 2: Comprehensive Performance Dashboard. **(a)** Phase-Jump Robustness (Scen. D): RA-EKF incurs zero threshold breaches (all  $N = 30$  runs). [cite: 942] **(b)** Ramp Tracking (Scen. B): RA-EKF eliminates lag via explicit  $\omega$  state. [cite: 935] **(c–d)** Multi-Event: stable vs. unstable estimators. [cite: 951] **(e)** Pareto frontier ( $\dagger$ PI-GRU excluded). [cite: 1091] **(f)** Radar profile. **(g)** Compliance heatmap: RA-EKF passes all thresholds under Composite Islanding ( $peak = 0.365$  Hz,  $T_{trip} = 0$ ). [cite: 1578] **(h)** Steady-state error (log scale, Scen. C). [cite: 1299]

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